

# Math

## 22 QUESTIONS

### DIRECTIONS

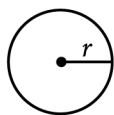
The questions in this section address a number of important math skills.  
Use of a calculator is permitted for all questions.

### NOTES

Unless otherwise indicated:

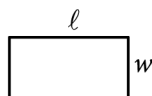
- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function  $f$  is the set of all real numbers  $x$  for which  $f(x)$  is a real number.

### REFERENCE

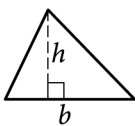


$$A = \pi r^2$$

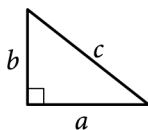
$$C = 2\pi r$$



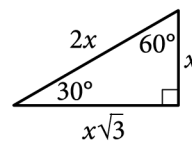
$$A = \ell w$$



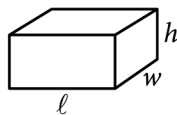
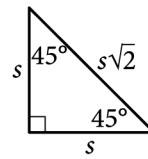
$$A = \frac{1}{2}bh$$



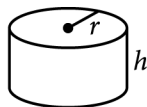
$$c^2 = a^2 + b^2$$



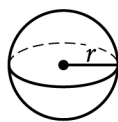
Special Right Triangles



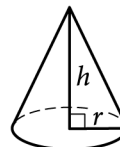
$$V = \ell wh$$



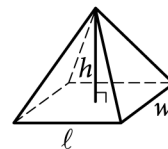
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is  $2\pi$ .

The sum of the measures in degrees of the angles of a triangle is 180.

**For multiple-choice questions**, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

**For student-produced response questions**, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as  $3\frac{1}{2}$ ), write it as an improper fraction ( $\frac{7}{2}$ ) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

Question ID: b988eeec

| Assessment | Test | Domain  | Skill            | Difficulty                               |
|------------|------|---------|------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard <div></div> <div></div> <div></div> |

Question

The functions  $f$  and  $g$  are defined as  $f(x) = \frac{1}{4}x - 9$  and  $g(x) = \frac{3}{4}x + 21$ . If the function  $h$  is defined as  $h(x) = f(x) + g(x)$ , what is the  $x$ -coordinate of the  $x$ -intercept of the graph of  $y = h(x)$  in the  $xy$ -plane?

Question ID: 93de3f84

| Assessment | Test | Domain                    | Skill           | Difficulty                                           |
|------------|------|---------------------------|-----------------|------------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard<br><div><div></div><div></div><div></div></div> |

### Question

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

**Answer**

- A. 11  
B. 22  
C. 44  
D. 66

## Question ID: 6011a3f8

| Assessment | Test | Domain        | Skill                                                                         | Difficulty    |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard<br>■■■■■ |

## Question

$$64x^2 + bx + 25 = 0$$

In the given equation,  $b$  is a constant. For which of the following values of  $b$  will the equation have more than one real solution?

## Answer

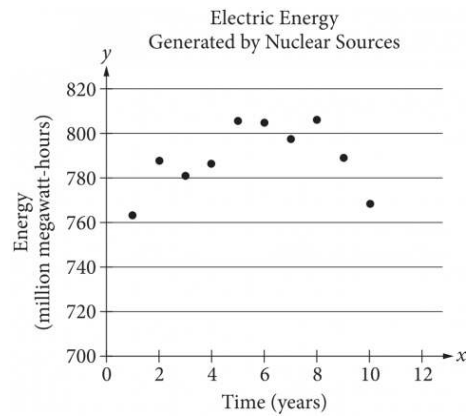
- A.  $-91$
- B.  $-80$
- C.  $5$
- D.  $40$

## Question ID: e821a26d

| Assessment | Test | Domain                            | Skill                                      | Difficulty |
|------------|------|-----------------------------------|--------------------------------------------|------------|
| SAT        | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard       |

## Question

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



Of the following equations, which best models the data in the scatterplot?

## Answer

- A.  $y = 1.674x^2 + 19.76x - 745.73$
- B.  $y = -1.674x^2 - 19.76x - 745.73$
- C.  $y = 1.674x^2 + 19.76x + 745.73$
- D.  $y = -1.674x^2 + 19.76x + 745.73$

Question ID: 771bd0ca

| Assessment | Test | Domain  | Skill                            | Difficulty                               |
|------------|------|---------|----------------------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard <div></div> <div></div> <div></div> |

Question

$5(t + 3) - 7(t + 3) = 38$

What value of *t* is the solution to the given equation?

Question ID: ca2235f6

| Assessment | Test | Domain                    | Skill   | Difficulty                                        |
|------------|------|---------------------------|---------|---------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div><div></div><div></div><div></div></div> |

**Question**

A circle has center  $O$ , and points  $A$  and  $B$  lie on the circle. The measure of arc  $AB$  is  $45^\circ$  and the length of arc  $AB$  is 3 inches. What is the circumference, in inches, of the circle?

- Answer**
- A. 3
  - B. 6
  - C. 9
  - D. 24

Question ID: de91a6fc

| Assessment | Test | Domain        | Skill               | Difficulty                                        |
|------------|------|---------------|---------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div><div></div><div></div><div></div></div> |

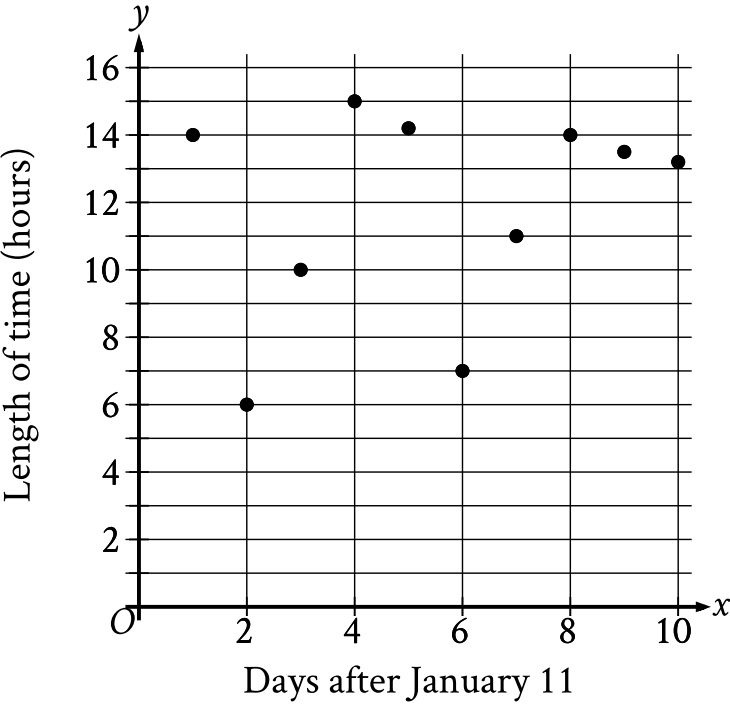
**Question**

A certain bowl was sold online for prices varying between **\$7** and **\$13**. The number of bowls sold can be modeled by a quadratic function based on the price, where **5** bowls were sold for **\$12** each and the maximum of **9** bowls were sold for **\$10** each. According to the model, what is the number of bowls that were sold for **\$8** each?

Question ID: 7b52985c

| Assessment | Test | Domain                            | Skill                                      | Difficulty                               |
|------------|------|-----------------------------------|--------------------------------------------|------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Two-variable data: Models and scatterplots | Hard <div></div> <div></div> <div></div> |

**Question**  
The scatterplot shows the relationship between the length of time  $y$ , in hours, a certain bird spent in flight and the number of days after January 11,  $x$ .



What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15?

Question ID: 0bd33265

| Assessment | Test | Domain  | Skill            | Difficulty                               |
|------------|------|---------|------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard <div></div> <div></div> <div></div> |

**Question**

The equation  $h = \frac{9(v-273.15)}{5} + 32$  gives the corresponding temperature  $h$ , in degrees Fahrenheit, of any substance that has a temperature of  $v$  kelvins, where  $v > 0$ . If a substance has a temperature of **467.33** degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

**Question ID: 179edb12**

| Assessment | Test | Domain                    | Skill           | Difficulty                                           |
|------------|------|---------------------------|-----------------|------------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard<br><div><div></div><div></div><div></div></div> |

### Question

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **18 cm**. The volume of this cone is  $n\pi \text{ cm}^3$ . What is the value of  $n$ ?

**Question ID: 371cbf6b**

| Assessment | Test | Domain        | Skill                  | Difficulty                                           |
|------------|------|---------------|------------------------|------------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

$$(ax+3)(5x^2-bx+4)=20x^3-9x^2-2x+12$$

The equation above is true for all  $x$ , where  $a$  and  $b$  are constants. What is the value of  $ab$ ?

**Answer**

- A. 18
- B. 20
- C. 24
- D. 40

## Question ID: aa43b41f

| Assessment | Test | Domain                            | Skill                                                                | Difficulty    |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard<br>■■■■■ |

**Question**

Near the end of a US cable news show, the host invited viewers to respond to a poll on the show's website that asked, "Do you support the new federal policy discussed during the show?" At the end of the show, the host reported that 28% responded "Yes," and 70% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of the population of the United States?

**Answer**

- A. The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B. Those who responded to the poll were not a random sample of the population of the United States.
- C. There were not 50% "Yes" responses and 50% "No" responses.
- D. The show did not allow viewers enough time to respond to the poll.

| Assessment | Test | Domain  | Skill            | Difficulty                                           |
|------------|------|---------|------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Linear functions | Hard<br><div><div></div><div></div><div></div></div> |

According to data provided by the US Department of Energy, the average price per gallon of regular gasoline in the United States from September 1, 2014, to December 1, 2014, is modeled by the function  $F$  defined below, where  $F(x)$  is the average price per gallon  $x$  months after September

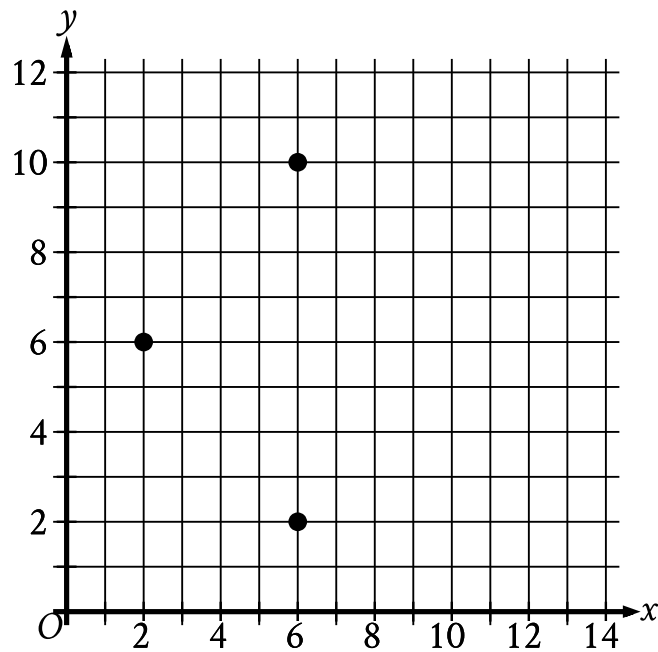
$$F(x) = 2.74 - 0.19(x - 3)$$

- A. The average monthly decrease in the price per gallon
- B. The difference in the average price per gallon from September 1, 2014, to December 1, 2014
- C. The average price per gallon on September 1, 2014
- D. The average price per gallon on December 1, 2014

**Question ID: b2528e6b**

| Assessment | Test | Domain                    | Skill           | Difficulty                                           |
|------------|------|---------------------------|-----------------|------------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard<br><div><div></div><div></div><div></div></div> |

### Question



The three points shown define a circle. The circumference of this circle is  $k\pi$ , where  $k$  is a constant. What is the value of  $k$ ?

## Question ID: b73ee6cf

| Assessment | Test | Domain        | Skill               | Difficulty                |
|------------|------|---------------|---------------------|---------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard<br>■■■■■ ■■■■■ ■■■■■ |

## Question

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years,  $t$ , it will take for the population of the town to reach 60,000 ?

## Answer

A.  $50,000 = 60,000(0.03)^t$

B.  $50,000 = 60,000(3)^t$

C.  $60,000 = 50,000(0.03)^t$

D.  $60,000 = 50,000(1.03)^t$

## Question ID: 7d68096f

| Assessment | Test | Domain                            | Skill                                                                | Difficulty    |
|------------|------|-----------------------------------|----------------------------------------------------------------------|---------------|
| SAT        | Math | Problem-Solving and Data Analysis | Evaluating statistical claims: Observational studies and experiments | Hard<br>■■■■■ |

## Question

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week. For the study, the organizer selected **55** teams at random from all trivia teams in a certain tournament. The table displays the information for the **40** teams in the sample that practiced for at least **3** hours per week.

| Hours practiced   | Number of points per round |                   |       |
|-------------------|----------------------------|-------------------|-------|
|                   | 6 to 13 points             | 14 or more points | Total |
| 3 to 5 hours      | 6                          | 4                 | 10    |
| More than 5 hours | 4                          | 26                | 30    |
| Total             | 10                         | 30                | 40    |

Which of the following is the largest population to which the results of the study can be generalized?

## Answer

- A. All trivia teams in the tournament that scored **14** or more points in the round
- B. The **55** trivia teams in the sample
- C. The **40** trivia teams in the sample that practiced for at least **3** hours per week
- D. All trivia teams in the tournament

Question ID: d1b66ae6

| Assessment | Test | Domain  | Skill                                            | Difficulty                                           |
|------------|------|---------|--------------------------------------------------|------------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard<br><div><div></div><div></div><div></div></div> |

### Question

$$-x + y = -3.5$$

$$x + 3y = 9.5$$

If  $(x, y)$  satisfies the system of equations above, what is the value of  $y$  ?

Question ID: f6ca90cc

| Assessment | Test | Domain                    | Skill           | Difficulty                                           |
|------------|------|---------------------------|-----------------|------------------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Area and volume | Hard<br><div><div></div><div></div><div></div></div> |

### Question

A right prism has a square base with side lengths of **10** inches. The height of the prism is **58** inches. What is the volume, in cubic inches, of the prism?

**Question ID: a0b4103e**

| Assessment | Test | Domain        | Skill                  | Difficulty                                           |
|------------|------|---------------|------------------------|------------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

The expression  $\frac{1}{3}x^2 - 2$  can be rewritten as  $\frac{1}{3}(x - k)(x + k)$ , where  $k$  is a positive constant. What is the value of  $k$ ?

**Answer**

- A. 2
- B. 6
- C.  $\sqrt{2}$
- D.  $\sqrt{6}$

**Question ID: 5acbd30**

| Assessment | Test | Domain        | Skill               | Difficulty                                           |
|------------|------|---------------|---------------------|------------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard<br><div><div></div><div></div><div></div></div> |

### Question

The function  $f$  is defined by  $f(x) = 56(0.19)^x$ . For any positive integer  $n$ , the value of  $f(n)$  is  $p\%$  less than the value of  $f(n - 1)$ . What is the value of  $p$ ?

**Answer**

- A. 19  
B. 44  
C. 56  
D. 81

Question ID: 84dd43f8

| Assessment | Test | Domain        | Skill               | Difficulty                               |
|------------|------|---------------|---------------------|------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear functions | Hard <div></div> <div></div> <div></div> |

Question

For the function  $f$ ,  $f(0) = 86$ , and for each increase in  $x$  by 1, the value of  $f(x)$  decreases by 80%. What is the value of  $f(2)$ ?

## Question ID: 9a1f1941

| Assessment | Test | Domain        | Skill                                                                         | Difficulty    |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard<br>■■■■■ |

## Question

$$|4x - 3| = -9$$

How many solutions does the given equation have?

## Answer

- A. Zero
- B. One
- C. Two
- D. Infinitely many